

Global patterns in plant–pollinator specialization

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Pollination is a key ecological process sustaining biodiversity and food security, yet global patterns of plant–pollinator specialization remain unresolved. Using a global dataset of quantitative plant–pollinator networks (>3,400 networks, >110,000 interactions), we show that specialization is latitudinally structured, but not as a single directional pattern. We find no consistent monotonic increase or decrease in specialization towards the Equator. Instead, network-level and pollinator specialization peak at low northern latitudes near the tropical–subtropical boundary, whereas plant specialization is hemispherically asymmetric. Climate is most often the best-supported predictor of specialization compared with latitude, biodiversity and environmental productivity. Network-level specialization declines with increasing mean annual temperature, with strong group-specific differences. Precipitation exerts nonlinear effects: plant specialization peaks at intermediate precipitation but is lower at the wettest sites, whereas pollinator specialization is lower at intermediate precipitation, although often in interaction with temperature. Pollinator groups diverge strongly, especially in contrasting precipitation-related patterns between birds and insects. These findings show that global plant–pollinator specialization is structured by latitude and climate in complex and taxon-specific ways, with implications for uneven responses of pollination networks to future climatic change.

The specialization of plant–pollinator interactions fundamentally influences pollination efficiency, plant reproduction and ecosystem functioning and stability. The extent to which species rely on limited partners shapes resource partitioning and coexistence, thereby structuring communities, affecting species persistence and driving evolutionary processes that generate and maintain biodiversity^{1–3}. Specialized pollination may also facilitate reproductive isolation and speciation by promoting niche differentiation and reducing competition, contributing to the exceptional diversity of plants and pollinators^{2,4}. Yet despite its ecological and evolutionary importance, broad-scale patterns in interaction specialization remain unresolved.

The latitudinal specialization gradient (LSG), a non-random latitudinal pattern in the degree of interaction specialization, remains a fundamentally unresolved question in ecology^{5,6}. Clarifying whether

the LSG exists, which shape it takes and what its drivers are is central to understanding the origin and maintenance of biodiversity gradients, community assembly and the evolution of species interactions^{7,8}. Influential hypotheses have often predicted strongly directional latitudinal patterns in specialization, for example^{9–11}, but the complex interplay of potential mechanisms, including climate variability, regional differences in evolutionary history or strong differences among interacting groups, may result in more complex non-random LSGs, rather than a single monotonic and symmetric gradient⁸. As specialization and species richness are theoretically coupled, the LSG is closely linked to the latitudinal diversity gradient, which describes increasing species richness towards the Equator^{1,9,11,12}. Testing whether the LSG exists, how it varies and what drives it is essential to understanding mechanisms that structure global biodiversity and ecosystem functioning and

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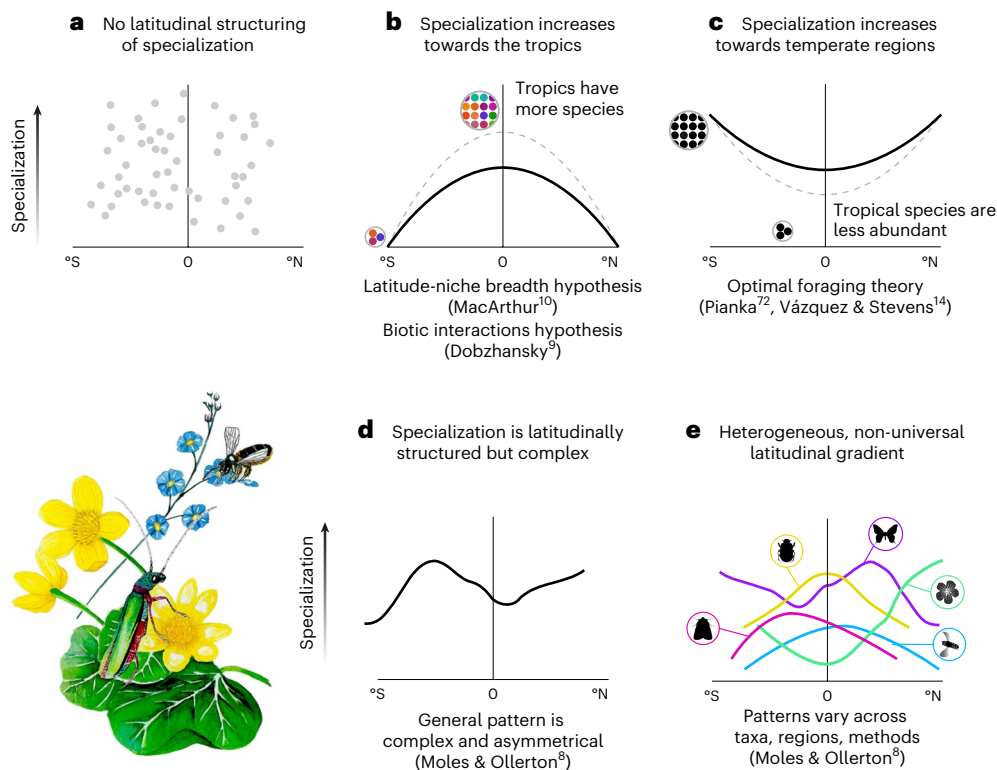


Fig. 1 | Hypothesized latitudinal patterns in interaction specialization.

Solid black lines show the predicted patterns; dashed lines illustrate gradients of the proposed mechanism underlying the specialization pattern. **a**, No detectable effects of latitude on specialization. **b**, Specialization increases towards species-rich and more stable tropical ecosystems under the latitude-niche breadth hypothesis¹⁰ and the biotic interactions hypothesis⁹, respectively. **c**, Specialization declines towards tropical ecosystems with more species which are less abundant under the optimal-foraging theory^{14,72}. **d**, Specialization

is significantly structured along the latitudinal gradient, but the general pattern is complex and non-directional towards the Equator⁸. **e**, Context-specific latitudinal specialization patterns varying among taxa, regions and/or methodological approaches⁸. Credit: Illustration by S.P.S. Silhouettes from PhyloPic (<https://www.phylopic.org/>) under a Creative Commons license: *Geranium maculatum*, *Luperina testacea*, *Papilio machaon*, *Eupeodes volucris* (CC0 1.0); *Onthophagus taurus* (CC BY-SA 3.0).

determine their responses to global change^{10,11,13}. Pollination networks are ideal for testing the LSG, because plant-pollinator interactions occur across all terrestrial biomes, encompass broad phylogenetic and functional diversity and represent mutualisms integral to community assembly and diversification^{7,11,14}.

Theoretical predictions and empirical evidence suggest several contrasting expectations for LSG in plant-pollinator interactions (Fig. 1). The null expectation predicts no detectable role of latitude in shaping specialization, and thus no LSG (Fig. 1a), in agreement with several empirical studies finding no systematic LSG patterns^{15–17}. When latitude does structure specialization, however, four broad LSG patterns are possible: increasing specialization towards the tropics; decreasing specialization towards the tropics; a general but complex nonlinear pattern that is not simply directional along latitude; or no general pattern because different taxa, regions or methodological approaches show different latitude-associated trends. Two prominent theoretical frameworks predict increasing specialization towards the tropics^{6,11} (Fig. 1b). The latitude-niche breadth hypothesis argues that strong competition and benign climates in tropical ecosystems cause ecological niches to narrow towards the Equator¹⁰, whereas the biotic interactions hypothesis links co-evolutionary specialization to the long-term climatic stability of tropical ecosystems⁹. Supporting empirical evidence for the increasing pollination specialization towards the tropics includes geographic patterns in plant-hummingbird networks¹⁸ and a global analysis of LSG in 54 pollination networks¹⁹. In direct contrast, the optimal-foraging theory predicts the reverse pattern of LSG, arguing that high species richness often coincides with lower local abundances of interacting species, causing tropical

species to generalize rather than specialize in order to obtain sufficient resources^{7,14} (Fig. 1c). This prediction is consistent with a global analysis of 58 pollination networks showing decreasing network-level specialization towards the tropics⁷. Finally, questioning any universal LSG, Moles and Ollerton⁸ argued that apparent latitudinal patterns in specialization are strongly context dependent and may vary with other confounding environmental drivers (such as elevation, climate or environmental productivity), interaction type, interacting taxa and ecosystem context (Fig. 1d).

Additionally, the conflicting evidence in existing plant-pollinator studies could be attributed to geographic, taxonomic and methodological biases⁸. Geographically, global analyses have been dominated by temperate networks from the Americas and Europe, leaving critical gaps elsewhere^{3,20} and have often overlooked regional differences, despite strong evidence for ecosystem-specific effects on specialization²¹. Taxonomically, the existing studies either have narrowed their focus to a single group (for example, bees²², hummingbirds¹⁸, cacti²³) or have combined groups indiscriminately (for example, refs. 15,19,24). The former may exaggerate, whereas the latter may obscure group-specific patterns shaped by contrasting diversity gradients, evolutionary histories and resource-use strategies^{25,26}. Methodologically, the use of non-standardized specialization metrics, uneven sampling effort and the mixing of complete community networks with taxon-constrained partial networks make cross-study comparisons unreliable^{20,25–28}. Overcoming these biases is essential for the rigorous testing of plant-pollinator LSG.

Here we ask what role latitude plays in the global structuring of plant-pollinator specialization and whether any global patterns in

specialization are better explained by climate, plant and pollinator diversity or environmental productivity. To address these questions, we assembled a comprehensive global dataset of quantitative plant–pollinator networks (Fig. 2), comprising 3,415 networks from 162 studies (see Supplementary Table 1 for the complete list) and covering 110,571 pairwise interactions among 5,343 pollinator and 6,126 plant species across all major terrestrial biomes and a broad latitudinal range (Fig. 3). We address key limitations of previous syntheses by standardizing sampling completeness (Extended Data Fig. 1), using two robust specialization metrics that quantify deviations from random partner choice based on interaction frequencies (one across the entire network (H_2') and the other averaged across species within a community (d') that allows separate analyses of particular functional groups) and accounting for methodological and regional differences. To separate biological patterns from artefacts caused by differences in taxonomic scope of network studies, we analyse latitudinal patterns separately for three network datasets differing in taxonomic resolution, from all networks pooled irrespective of taxonomic scope, to community-wide full networks and networks resolved by pollinator functional groups. Crucially, we analyse plants and pollinator functional groups separately, testing predictions that latitudinal patterns reflect their ecological and evolutionary differences. By combining unprecedented global coverage with a framework that overcomes key geographic, taxonomic and methodological constraints and explicitly analyses functional-group-specific patterns, our study advances understanding of biodiversity gradients and the architecture of mutualistic networks.

Results

LSG

We first asked whether latitude plays an important role in structuring interaction specialization at all: that is, whether any LSG can be detected (Fig. 1a and Fig. 2j,k). Using generalized additive mixed models (GAMMs) and hierarchical GAMMs (HGAMs) on the three network datasets (all networks, full networks and functional-group networks), we modelled the three specialization metrics (measuring deviation from random interaction partner choice at the network-level H_2' and community-mean species-level d'_{plants} and $d'_{\text{pollinators}}$) against latitude while accounting for methodological biases known to affect patterns reported in previous studies (Fig. 3, Supplementary Fig. 1 and Supplementary Tables 4 and 5). Network descriptors that significantly influenced latitudinal patterns were retained as random effects in the final models, with geographic region and habitat openness doing so most frequently (Supplementary Fig. 1 and Supplementary Tables 4 and 5). We then visually compared the fitted relationships among models with a significant latitude effect to determine which of the possible LSGs best described patterns of each metric and dataset: increasing or decreasing specialization towards the Equator (Fig. 1b,c), a general but complex nonlinear pattern (Fig. 1d) or heterogeneous functional-group-specific latitudinal patterns (Fig. 1e).

In the all-networks dataset, latitude significantly predicted all three specialization metrics, but the fitted relationships differed strongly among them (Extended Data Fig. 2). Moreover, the H_2' and

d'_{plants} patterns contrasted with results from the better-resolved datasets (see below), whereas $d'_{\text{pollinators}}$ more closely resembled them by showing highest values at northern low latitudes. Because these discrepancies likely result from heterogeneity in taxonomic scope when all networks are combined, we do not interpret these results further.

Across the full-networks and functional-group-networks datasets, latitude significantly structured specialization in four of the five metric–dataset combinations (Fig. 4 and Supplementary Table 5); only H_2' in the full-networks dataset showed no detectable latitudinal structuring (Fig. 4a). In the remaining cases, the fitted relationships were most consistent either with a complex nonlinear LSG or with heterogeneous functional-group-specific LSGs; none provided clear overall support for a monotonic increase or decrease in specialization towards the Equator (Fig. 1). Where latitude did matter, the strongest structure generally emerged at low northern latitudes, repeatedly peaking around 20–30° N (Fig. 4).

In the full-networks dataset, latitude explained relatively little variation ($R^2 \leq 0.09$). The latitudinal pattern of $d'_{\text{pollinators}}$ was best described as a complex nonlinear LSG, showing highest values at low northern latitudes (~20–25° N), followed by a marked decline and a secondary rise at higher northern latitudes (Fig. 4b). Most pollinator groups did not deviate significantly from this overall trend, although moths showed an additional peak in the southern tropics. In contrast, d'_{plants} showed only a weakly supported ($R^2 = 0.01$) hemispherically asymmetric latitudinal pattern, increasing linearly towards the Equator in the Southern Hemisphere and showing the opposite linear decrease towards the Equator in the Northern Hemisphere (Fig. 4c).

In the functional-group-networks dataset, our models revealed stronger taxon-specific differences in the role of latitude in structuring specialization, together with a stronger relationship between latitude and specialization ($R^2 = 0.21$ – 0.25 ; Supplementary Table 5 and Fig. 4d,e). The general relationship between H_2' and latitude was broadly hump-shaped, although hemispherically asymmetric, peaking at low northern latitudes (~20–25° N), declining slightly towards the Equator and declining strongly towards higher latitudes in both hemispheres, with a flat pattern at the highest northern latitudes (Fig. 4d). Most functional groups that significantly deviated from this trend (bees, butterflies, hoverflies and moths) showed broadly similar latitudinal specialization patterns, whereas birds showed an S-curved pattern with a dip in the southern tropics and a peak in the northern tropics (Fig. 4d). These specialization patterns are therefore most consistent with a complex, nonlinear LSG. By contrast, $d'_{\text{pollinators}}$ showed patterns most consistent with heterogeneous functional-group-specific LSGs (Fig. 4e). The overall fitted curve was bimodal, with highest values close to both tropics, a dip around the Equator and a steep decline towards high northern latitudes. However, most functional groups (bees, birds, butterflies, hoverflies and moths) showed strongly different patterns, often with multiple local maxima and minima (Fig. 4e). Among these, hoverflies most closely matched the expected increase in specialization towards the Equator, birds showed the opposite pattern, whereas moths showed especially irregular latitude-associated changes with several maxima and minima (Fig. 4e).

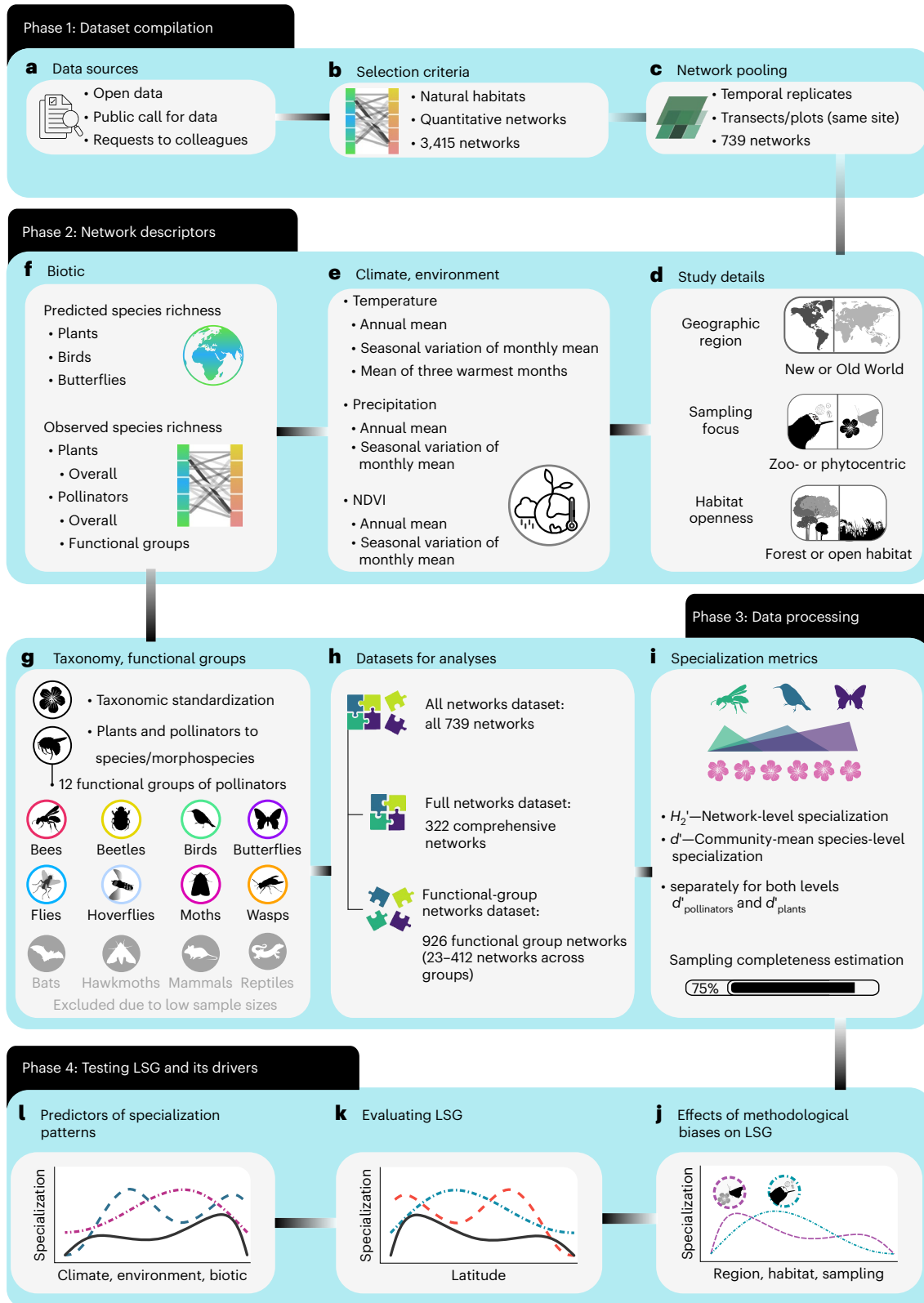
Fig. 2 | Conceptual workflow for analysing the LSG in plant–pollinator interactions. a–c, Phase 1: compilation of available plant–pollinator networks (a), application of predefined selection criteria (b) and pooling of spatiotemporal network replicates from the same site to minimize non-focal variation (c). **d–f, Phase 2:** characterization of networks with 28 descriptors, including study details (d), climate and environment (e) and biotic (f) (Supplementary Table 2). **g–i, Phase 3:** taxonomic standardization and assignment of pollinators to functional groups (g) (grey icons indicate groups excluded from functional-group analyses due to data insufficiency), followed by construction of three complementary datasets (h) and calculation of three specialization metrics and sampling completeness for each sampling network (i). **j–l, Phase 4:** multistage modelling of the LSG, including evaluation of potential methodological biases

(j), fitting latitudinal models with significant biases as random effects (k) and testing climatic, environmental and biotic predictors to identify the predictors of specialization patterns (l). See Methods for details. Credit: Icons from Noun Project, reproduced under a Creative Commons licence CCBY3.0; search, rivercon; world globe, Rahe; world map, Donata Bologna. Credit: Silhouettes from PhyloPic (<https://www.phylopic.org/>) under a Creative Commons license: *Microchera albocoronata*, *Luperina testacea*, *Geranium maculatum*, *Bombus terrestris*, *Acanthorhynchus tenuirostris*; *Papilio machaon*, Tachinidae, *Eupeodes volucris*, *Vespula germanica*, *Apodemus sylvaticus*, *Lepidodactylus lugubris* (CC0 1.0); *Aglais urticae*, *Darapsa myron*, *Desmodus rotundus* (CC BY 3.0); *Halictus rubicundus* (CC BY 4.0); *Onthophagus taurus* (CC BY-SA 3.0). Search, world globe and world map icons from Noun Project.

Predictors of specialization patterns

We next tested how forward-selected climatic (mean annual temperature (MAT), mean annual precipitation (MAP) and temperature seasonality (standard deviation of mean monthly temperature (SD_MMT))), environmental (normalized difference vegetation index (NDVI) and standard deviation of monthly NDVI (SD_NDVI)) and biotic (12 measures of predicted and observed species richness of plants and pollinators) predictors (see latitudinal distributions in Extended Data Fig. 3)

compared with latitude in explaining global variation in specialization (H_2' , $d'_{\text{pollinators}}$ and d'_{plants}). Using the same GAMM/HGAM framework (Fig. 2i), we compared individual predictors (and the MAT $\times$ MAP interaction) among each other and against the corresponding latitudinal models, using a combination of model Akaike information criterion (AIC) and marginal R^2 (Fig. 5 and Supplementary Tables 6 and 7). All three potential methodological biases (geographic region, sampling focus and habitat openness) showed significant effects in some models



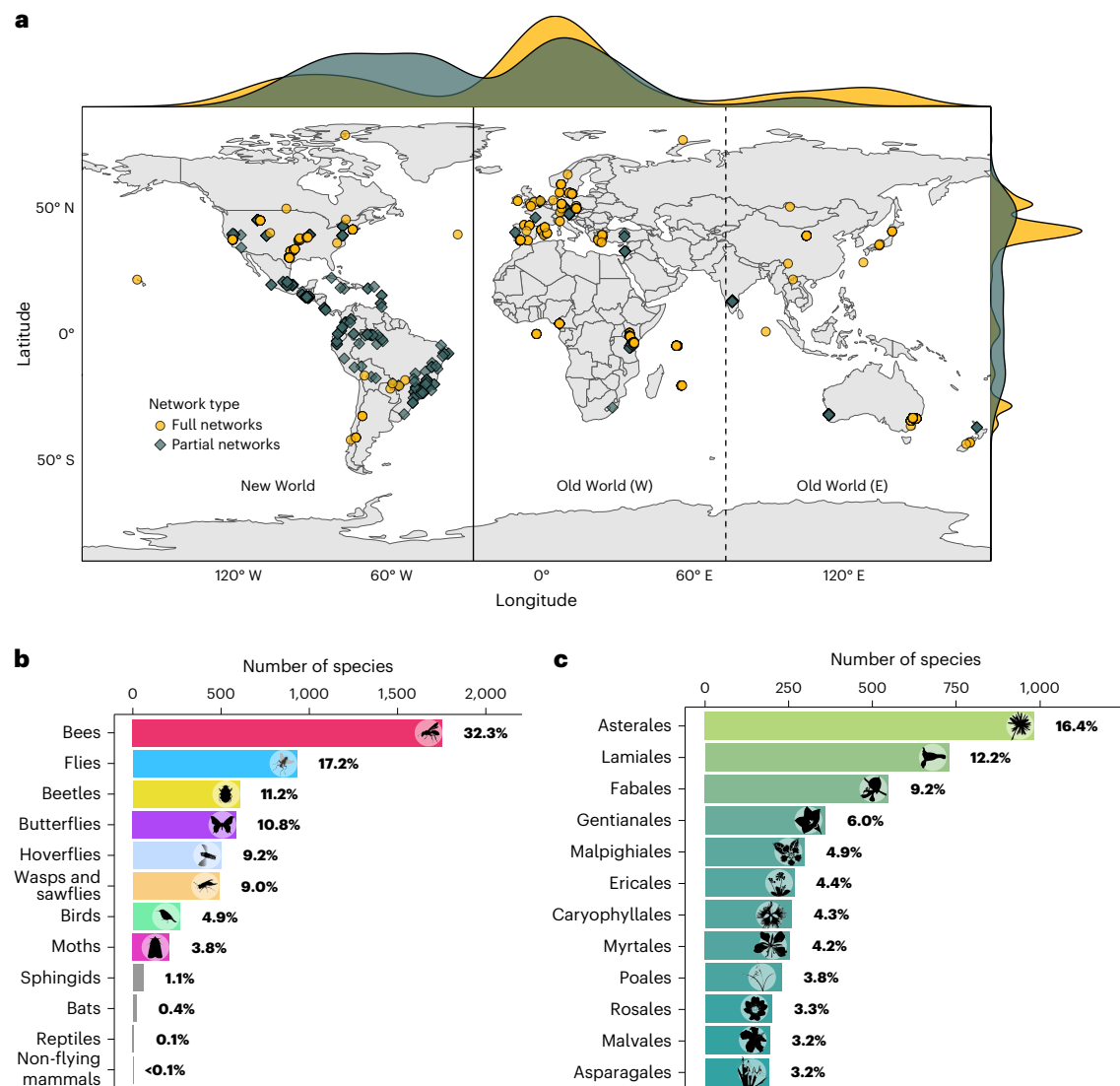


Fig. 3 | Global coverage and taxonomic composition of the analysed plant-pollinator networks. (a) Geographic distribution of networks, grouped into the New World and Old World. Symbols denote network completeness: taxonomically restricted (partial) and unrestricted (full). Marginal density plots show the distributions of network locations by their completeness along latitude (right) and longitude (top). (b) Pollinator composition by functional group, with bars indicating their percentage out of all pollinator species in the dataset. (c) Plant composition, showing the 12 most speciose orders and their

percentages in the dataset. Credit: Silhouettes from PhyloPic (<https://www.phylopic.org/>) under a Creative Commons license: *Halictus rubicundus*, *Dianthus hyssopifolius* (CC BY 4.0); *Onthophagus taurus* (CC BY-SA 3.0); *Acanthorhynchus tenuirostris*, *Papilio machaon*, Tachinidae, *Eupeodes volucris*, *Galanthus nivalis*, *Luperina testacea*, *Vespula germanica*, *Crepis paludosa*, *Penstemon buckleyi*, Leguminosae, *Primula veris*, *Chamaenerion angustifolium*, *Oryza sativa*, *Dryas integrifolia*, *Hibiscus* (CC0 1.0).

(Supplementary Table 4) and were therefore retained as random effects where applicable.

The best-supported models retained climatic predictors, although ΔAIC among some models with other predictors was sometimes relatively small. MAT best explained H_2' in the full-networks dataset (Fig. 5a). The MAT $\times$ MAP interaction performed best for $d'_{\text{pollinators}}$ in both datasets (Fig. 5b,c,f,g) and for d'_{plants} in the full-networks dataset (Fig. 5d,h). In the functional-group-networks dataset, MAP and latitude performed comparably for H_2' (Figs. 4d and 5e and Supplementary Table 7). Environmental productivity and biodiversity models often explained less variation and ranked below both latitudinal and climatic models (Supplementary Table 7). In the full-networks dataset, climatic predictors explained substantially more variance with larger marginal effects than latitude for H_2' and d'_{plants} , whereas their advantage over latitude was modest for $d'_{\text{pollinators}}$ (Figs. 4 and 5 and Supplementary Table 7).

Climate effects on specialization were frequently nonlinear and differed among metrics and datasets. In the full-networks dataset, H_2' showed a nonlinear but overall declining relationship with MAT (Fig. 5a and Supplementary Table 7). For $d'_{\text{pollinators}}$, the response to MAT was weak and linear (Fig. 5b), whereas the response to MAP was strongly nonlinear (Fig. 5c). $d'_{\text{pollinators}}$ dropped sharply at intermediate precipitation, and confidence intervals became increasingly wide at the wettest sites. Three functional groups deviated from the overall trend, especially for MAP. Hoverfly specialization increased linearly, moths showed a hump-shaped response, and beetles showed a minimum at intermediate MAP. d'_{plants} showed an S-shaped response to MAT, peaking just below 0 °C and dipping near 12 °C (Fig. 5d), and a near-hump-shaped response to MAP, peaking at intermediate precipitation (~3,000 mm annually), although only a few data points originated from wetter regions (Fig. 5h).

In the functional-group-networks dataset, H_2' showed no significant overall relationship with MAP, but five functional groups

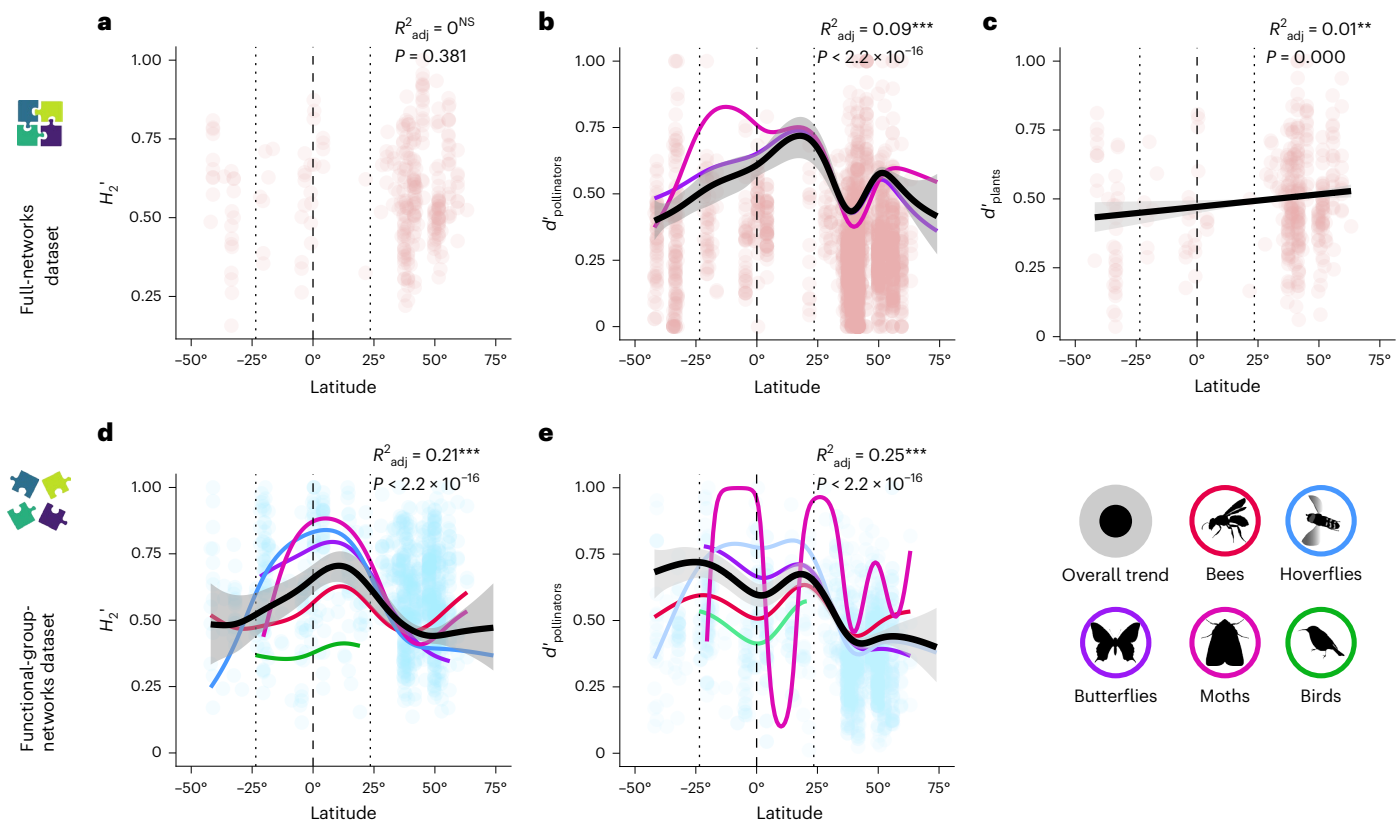


Fig. 4 | Latitudinal patterns in specialization of plant–pollinator interactions across datasets and metrics. **a–e**, Rows show datasets: full networks (**a–c**), functional-group networks (**d,e**). Columns show specialization metrics: network-level H_2' (**a,d**) and community-mean species-level separated for pollinators ($d'_{pollinators}$) (**b,e**) and plants (d'_{plants}) (**c**); not calculated for the functional-group-networks dataset. A vertical dashed line marks the Equator (0°), and vertical dotted lines mark the Tropics of Capricorn (-23.5°) and Cancer (23.5°). Curves are GAMM/HGAM fitted mean response smooths: black curves indicate overall latitudinal trends with 95% confidence intervals (grey shading), coloured curves

represent significant functional-group-specific trends (see legend). Coloured points are raw data. Curves are drawn only when the smooth term for latitude is significant; otherwise only points are shown. Adjusted R^2 and exact P values for the overall trend are reported in each panel (see Supplementary Table 5 for full model summaries). We assessed the significance of smooth terms using two-sided approximate F -tests without correction for multiple comparisons. Credit: Silhouettes from PhyloPic (<https://www.phylopic.org/>) under a Creative Commons license: *Halictus rubicundus* (CC BY 4.0); *Acanthorhynchus tenuirostris*, *Papilio machaon*, *Eupeodes volucris*, *Luperina testacea* (CC0 1.0).

diverged markedly (Fig. 5e): hoverflies, bees, beetles and moths all became more specialized with increasing precipitation, although moths showed a shallow minimum at $-2,200$ mm, whereas birds showed the opposite trend. $d'_{pollinators}$ increased linearly with MAT overall and showed a nonlinear response, with a sharp dip at intermediate MAP ($-3,000$ mm annually; Fig. 5f,g). Again, functional groups diverged: bees and beetles largely followed the overall MAT trend in specialization, bee specialization additionally peaked at intermediate precipitation ($-2,500$ mm MAP), and beetles became less specialized with increasing MAP, whereas bird specialization declined linearly with both climate predictors (Fig. 5f,g).

Discussion

Our global analysis of >3,400 quantitative plant–pollinator networks provides comprehensive evidence that plant–pollinator specialization is latitudinally structured, but the gradient is neither simple nor universal. Across the better-resolved datasets, latitude significantly structured specialization in most specialization metric–dataset combinations, yet none of these relationships supported a consistent monotonic increase or decrease in specialization towards the Equator. Instead, the detected patterns were best described either as complex nonlinear LSGs or as heterogeneous taxon-specific LSGs. Network level and pollinator specialization most often peaked at low northern latitudes, near the northern tropical–subtropical boundary, whereas plant specialization showed only a weak but strongly hemispherically

asymmetric pattern. Thus, our results reject both simple directional expectations of the LSG and instead support a more context-dependent view in which latitude matters, but its effect depends on metric, hemisphere and pollinator functional group.

None of the main theoretical frameworks fully explains these global patterns in plant–pollinator specialization. The latitude–niche breadth¹⁰ and biotic interactions hypotheses⁹ predict increasing specialization towards the Equator¹¹, whereas optimal-foraging theory predicts the opposite trend^{7,14}. Our results, however, support neither prediction as a general explanation. The clearest specialization peaks were displaced from the Equator to low northern latitudes, inconsistent with both classic LSG predictions (Fig. 1). The additional declines in both specialization metrics towards the highest latitudes may reflect increasing physiological constraints in extreme climates^{6,29}. Moreover, specialization in plants and several pollinator groups showed even more divergent or irregular patterns. At the same time, however, two group-specific trends partly resembled the two expected directional LSGs, including hoverflies showing the clearest tendency towards increasing specialization near the Equator and birds showing the opposite pattern. These contrasts suggest that mechanisms such as competition, resource availability, species abundances and physiological constraints probably operate differently across plant and pollinator groups⁸. Although neither species richness nor environmental productivity (a potential proxy for resource availability^{12,30}) outperformed climate predictors in our models, their contribution cannot be excluded,

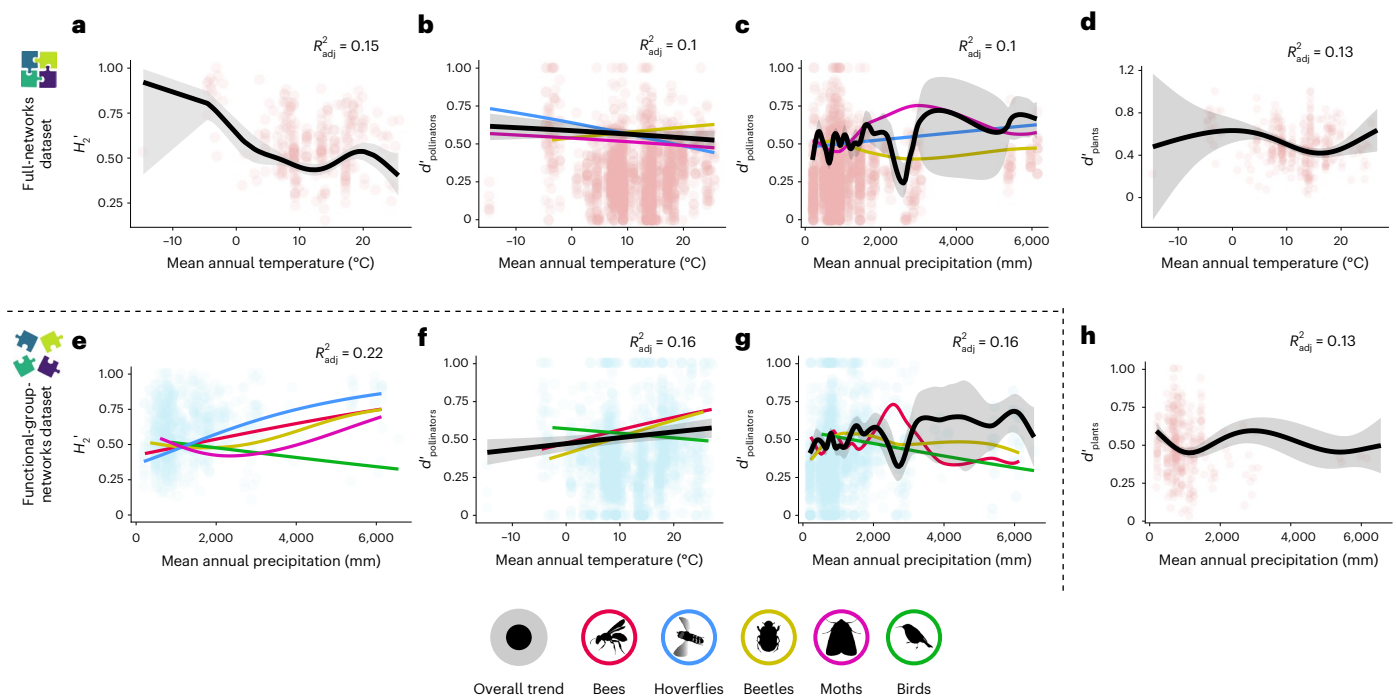


Fig. 5 | Best-supported climatic predictor models for plant–pollinator specialization across datasets and metrics. Panels show, for each metric–dataset combination, the predictors from the best-supported GAMM/HGAM model (lowest ΔAIC ; Supplementary Table 7). **a–h**, Results for the full-networks dataset (**a–d,h**) and the functional-group-networks dataset (**e–g**). Columns present specialization metrics: network-level H_2' (**a,e**) and community-mean species-level separated for pollinators ($d'_{\text{pollinators}}$) (**b,c,f,g**) and plants (d'_{plants}) (**d,h**); not calculated for the functional-group-networks dataset. Curves are

GAMM/HGAM fitted mean response smooths: black curves indicate significant overall predictor trends with 95% confidence intervals (grey shading), coloured curves represent significant functional-group-specific trends (see legend). Coloured points are raw data. Adjusted R^2 is reported in each panel; model details are in Supplementary Table 7. Credit: Silhouettes from PhyloPic (<https://www.phylopic.org/>) under a Creative Commons license: *Halictus rubicundus* (CC BY 4.0); *Onthophagus taurus* (CC BY-SA 3.0); *Acanthorhynchus tenuirostris*, *Eupeodes volucris*, *Luperina testacea* (CC0 1.0).

given the modest explanatory power of most predictors^{6,10,11,14}. Overall, with the current knowledge, the observed LSG should be interpreted primarily as a robust macroecological pattern that motivates more mechanistic tests, rather than as support for a single existing theory.

Critically, our findings are more robust than previous syntheses not only because of dataset size but also because we explicitly accounted for methodological biases (for example, refs. 8,31) and analysed different pollination network types separately. Pooling all networks irrespective of taxonomic scope produced highly contrasting patterns with very little explained variation (all-networks dataset; Extended Data Fig. 2), suggesting that indiscriminate combining of complete community networks with taxon-constrained partial networks can obscure or distort latitudinal signals. In contrast, analyses of full networks and especially functional-group networks revealed more interpretable patterns, with stronger explanatory power in the functional-group dataset. This supports the concern that methodological heterogeneity has contributed to conflicting conclusions in previous syntheses⁸. It also agrees with the broader expectation that any LSG in plant–pollinator interactions should be group- and context-dependent rather than universal⁸. Such heterogeneity may partly reflect functional-group differences in biogeography and ecological roles, including subtropical peaks in bee species richness³² and the increasing importance of flies in high-latitude pollination networks³³. Together, these results help reconcile why previous studies have reported increasing^{18,19}, decreasing⁷ or absent^{15–17} latitudinal patterns in pollination specialization.

Climatic predictors were mostly the best-supported models for plant–pollinator specialization compared to biodiversity, productivity and even latitude alone, but their interpretation requires caution. Because temperature strongly covaries with latitude

(Extended Data Fig. 4), and because candidate models differed in their retained random-effect structures, our model comparisons identify relative predictive performance rather than independent causal drivers. Within these limits, climate explained several patterns that can hardly be related to species richness or environmental productivity alone. The general decline of network-level specialization in the full-networks dataset with MAT contrasts with expectations that warmer and less seasonal environments should promote stronger specialization in plant–pollinator interactions^{7,11,34,35}. One possible explanation is that warmer climates extend pollinator activity and flowering windows within days and across seasons, increasing temporal overlap among potential partners and opportunities to switch partners and thus potentially promoting generalization^{36–38}. Conversely, colder or more seasonal climates may compress flowering and foraging into shorter windows, promoting tighter temporal matching and higher specialization, especially at the community level^{39,40}. Precipitation effects were more strongly nonlinear and differed among metrics and functional groups. Moderate rainfall may support flowering and foraging, but frequent or heavy rainfall can suppress flight activity⁴¹ and reduce floral rewards through nectar dilution or pollen wash-off^{42,43}. These mechanisms could help explain why rainfall was associated with contrasting responses of plants, pollinators and functional groups.

Differences among pollinator functional groups indicate that the LSG may arise from group-specific traits related to climate, as physiology and foraging ecology determine how strongly temperature and rainfall constrain activity and resource tracking^{44,45}. Several insect groups substantially contributed to the northern low-latitude peak in specialization, whereas birds showed weaker latitudinal structuring but a clearer decline in specialization with increasing precipitation. Such contrasts may reflect differences in thermal physiology and foraging

flexibility: insect activity and foraging efficiency are often strongly constrained by temperature and rainfall^{13,39}, whereas mostly tropical nectarivorous birds may respond more flexibly to year-round available but phenologically shifting resources⁴⁶. Weak or absent climatic trends in specialization of some insect groups, including flies and wasps, may reflect their heterogeneous diets and diverse life histories, although some clades are highly selective nectar or pollen feeders^{3,47}.

Our results suggest that climate is an important predictor of global plant–pollinator specialization patterns, although the climatic associations differ among functional groups. Because these associations are correlational, extrapolating them to future climates requires caution. Nevertheless, if they reflect underlying mechanisms, continued warming and altered precipitation regimes may reorganize pollination networks unevenly across groups and regions. Warming may favour more generalized network structure, consistent with the decline in network-level specialization with increasing MAT, although functional groups are likely to respond differently. Shifts in rainfall intensity and seasonality may disproportionately disrupt specialized interactions, which are particularly vulnerable to phenological mismatches and partner loss^{48–50}, with consequences for ecosystem stability that remain difficult to predict. Such effects should vary geographically and taxonomically, depending on baseline climates, the strength of climatic structuring and group-specific climate relationships. Overall, our results caution that climate change is unlikely to affect all plant–pollinator networks uniformly, emphasizing the importance of accounting for functional-group differences and regional climate baselines when assessing future risks to pollination services and biodiversity.

Methods

All analyses, including data preparation, were performed using R 4.5.0 (ref. 51).

Dataset compilation

We compiled a global dataset of plant–pollinator interaction networks (Fig. 1a) through a systematic search of Web of Science and Google Scholar between January and May 2021 (and we extensively monitored for new suitable studies until September 2024). Search strings combined terms ‘pollin*’, ‘network*’, ‘flower visit*’, ‘plant*’. We manually screened all retrieved publications for relevance and examined their cited and citing references. The dataset was supplemented with publicly available datasets from the Mangal Interaction Database⁵², the Web of Life database⁵³ and a smaller set of data from the currently non-public LifeWebs project⁵⁴. To maximize geographic and methodological coverage, we also contacted authors of studies without open data and requested both published and unpublished datasets from the international community of pollination ecologists (Supplementary Table 1).

Networks were included if they met the following criteria (Fig. 2b): originated from natural or seminatural ecosystems (that is, excluding agricultural, urban and other anthropogenic habitats), contained a minimum of five plant and five pollinator species, provided directly sampled interaction data (that is, excluding compilations over broad spatial or temporal scales) and reported quantitative interaction strengths (that is, excluding binary datasets). Coordinates, elevation, sampling details and other metadata were extracted from publications and/or databases when available. When essential information was missing, we contacted the original authors and excluded studies that could not be completed. Because direct measures of pollination efficiency were rarely available, we treated all floral visitors as potential pollinators and all flower visits as pollination interactions, which is standard practice in large-scale pollination network analyses. Our compiled dataset contained 3,415 quantitative networks from 162 studies, spanning 43.60° S to 81.00° N in latitude and 1–4,241 m in elevation, representing virtually all terrestrial biogeographic regions and major ecosystem types (Fig. 3a and Supplementary Table 1).

To reduce variation irrelevant to our aims, we pooled temporally replicated networks sampled at the same sites and merged networks representing small-scale spatial units (such as transects or plots) within the same study site (Fig. 2c). However, networks from studies explicitly examining elevational or environmental gradients were retained separately, as were networks from different publications to account for methodological differences. After these steps, our dataset contained 739 unique networks.

Network descriptors

Each plant–pollinator network was characterized using 28 descriptors (Fig. 2d–f) grouped into four categories: study and methodological details, geographic information, environmental conditions and local species richness (Supplementary Table 2; latitudinal trends in Extended Data Fig. 3).

Study and methodological details (Fig. 2d) included the source of network data, rationale for splitting datasets into separate networks (for example distinct sites, elevations, habitats or vertical strata), sampling design and field methods, sampling focus (zoocentric or phytocentric), network completeness (full or partial networks) and lower-level (plants) and upper-level (pollinators) constraints (specified if sampling was taxonomically biased). Geographic information comprised country, geographic coordinates, absolute latitude, elevation and region (New World or Old World; Fig. 2d). Environmental conditions included habitat openness (forest or open habitat) and seven additional climatic and environmental variables (Fig. 2e) extracted from geographic coordinates with the terra package⁵⁵. These climatic variables included MAT (°C), SD_MMT (±°C), mean temperature of the three warmest months (°C), MAP (mm) and the standard deviation of mean monthly precipitation (±mm)⁵⁶. Environmental productivity was quantified as annual mean NDVI⁵⁷ and SD_NDVI. Local species-richness descriptors (Fig. 2f) included predicted local species richness for plants⁵⁸, birds⁵⁹, nectarivorous birds (range maps from BirdLife, filtered using an unpublished species list by two of the authors (B.D. and J.O.) and butterflies⁶⁰, retrieved using the raster package⁶¹. In addition, we included observed species richness directly recorded in the original network data: plants, pollinators and total species richness, as well as richness of individual functional pollinator groups (see Supplementary Table 2 for definitions).

Taxonomy and functional groups

We standardized the taxonomy and nomenclature for plants and pollinators across all studies (Fig. 2g) using the taxize package⁶² with the GBIF database⁶³. Species names were verified by exact matching; unmatched or partially matched (‘fuzzy’) species names were corrected to fix typos and rematched. Taxa identified to morphospecies were retained as unique entities (called species hereinafter) within individual datasets but were not comparable across different studies.

Pollinator species were assigned to functional groups reflecting distinct floral resource use and ecological roles, following commonly accepted pollination syndromes⁴⁷: bees, wasps (including sawflies), butterflies, hawkmoths, settling moths, beetles, birds, bats, non-flying mammals and reptiles (Fig. 2g). Flies were further separated into hoverflies, known for their relatively specialized floral associations and high dataset representation, and all other flies, without further subdivision of specialized fly groups due to low abundance and generally coarse taxonomic identification in our data. Other floral visitors (such as spiders, ants orthopterans, hemipterans) were excluded as mostly accidental floral visitors and/or unimportant pollinators (although some species can play an important role in unique pollination systems⁶⁴). Plants were not subdivided into functional groups.

After taxonomic standardization and restricting networks to the selected pollinator functional groups, the dataset comprised 110,571 pairwise interactions among 5,343 pollinator species and 6,126 plant species, spanning a broad diversity of pollination systems (Fig. 3b,c).

Datasets for analyses

To reflect the dataset heterogeneity and robustly test latitudinal specialization patterns, we created three distinct data subsets for further analyses (Fig. 2h).

All-networks dataset (739 networks). This dataset included all quantitative networks after the above-described filtering and merging, regardless of original taxonomic scope or sampling methods. This allowed evaluation of global specialization patterns analogous to previous syntheses not correcting for methodological or taxonomic differences.

Full-networks dataset (322 networks). This dataset included only networks covering plant–pollinator communities without intentional taxonomic focus (that is, not necessarily sampling all potential interactions in the community, which is usually impossible even with substantial effort). We excluded partial networks explicitly restricted to plant or pollinator groups, such as hummingbird-visited flowers or bee–plant interactions. This dataset provides more robust insights into community-level specialization patterns by reducing bias from taxonomic sampling scope.

Functional-group-networks dataset (926 networks, varying by functional group). To test within-group specialization patterns, we used networks originally sampled with a taxonomic bias or subnetworks extracted from entire-community networks. These subnetworks were created separately for each functional pollinator group, retaining only those with ≥ 5 plant and ≥ 5 pollinator species (Supplementary Table 3). Due to insufficient data, reptiles, bats, non-flying mammals and hawk-moths were excluded from analyses of this dataset (Fig. 2g).

Specialization metrics

To comprehensively characterize specialization and allow reliable comparisons across networks of different size, we used two complementary size-independent metrics capturing network-level and species-level specialization (Fig. 2i), using the bipartite package⁶⁵.

Network-level specialization H_2' . This metric quantifies the deviation of observed interaction frequencies from random partner choice, reflecting overall interaction specialization⁶⁶. Values range from 0 (complete generalization; interactions randomly distributed) to 1 (maximum specialization; interactions highly restricted). This metric is standardized to account for variation in network size and species frequencies⁶⁶.

Community-mean species-level specialization d' . This metric represents the average specialization of individual species within networks, allowing separation for particular functional groups. Analogous to H_2' , this metric quantifies deviation of species from random interactions, ranging from 0 (generalist) to 1 (specialist)⁶⁶. Although d' is correlated with H_2' when calculated for all species in the network⁶⁶, it allows specialization quantification for a part of species, such as plants, pollinators or their functional groups. We calculated mean d' separately for pollinators ($d'_{\text{pollinators}}$) and plants (d'_{plants}). $d'_{\text{pollinators}}$ was calculated across all three datasets and further partitioned by functional group. d'_{plants} was calculated for the all-networks and full-networks datasets but not for the functional-group-networks dataset, to avoid biases introduced by restricting networks to a single pollinator group.

Analyses of specialization patterns

We fitted separate models for each of the three specialization metrics (H_2' , $d'_{\text{pollinators}}$ and d'_{plants}) using a beta distribution bounded between 0 and 1. To minimize sampling bias⁶⁷, all analyses were weighted by network-specific sampling completeness. This was estimated as the proportion of observed to total interactions (completeness at order $q = 0$) using rarefaction and extrapolation implemented in the

Completeness.link function from the iNEXT package⁶⁸. Networks were generally well sampled, with 75% exceeding 0.75 completeness, without any obvious latitudinal bias, except at the northernmost latitudes ($>65^\circ \text{N}$; Extended Data Fig. 1). The estimated completeness value of each network was directly used as its model weight, ensuring that each network contributed appropriately to the analyses. To assess the robustness of this approach, we repeated the full latitude–specialization analytical workflow using two alternative treatments of sampling completeness. First, we repeated analyses using only networks with sampling completeness >0.7 in each dataset. Second, we included sampling completeness as a parametric covariate, instead of a weighting term. These alternative approaches produced latitudinal model outputs that were generally very similar to the primary weighted analyses, and we therefore retained weighting by sampling completeness as the main approach while reporting the alternative results in Supplementary Tables 8 and 9 and Supplementary Figures 3 and 4.

To analyse patterns in interaction specialization, we constructed eight model sets for each of the three specialization metrics across the three datasets (with d'_{plants} not calculated for the functional-group-networks dataset), employing two complementary modelling approaches. GAMMs⁶⁹ were applied to H_2' and d'_{plants} in the all-networks and full-networks datasets to capture overall patterns, as GAMMs allow flexible nonlinear relationships without assuming a predefined shape. To incorporate functional-group-specific responses, we fitted HGAMs to H_2' in the functional-group-networks dataset and to $d'_{\text{pollinators}}$ in all three datasets, as HGAMs extend GAMMs by modelling both global and group-specific responses to predictors. Following ref. 69, we constructed five HGAM types: global effect model (single common response across functional groups); group-specific common-shape model (functional groups' responses differ in magnitude but share shape); group-specific different-shape model (each group has a distinct response curve); and two combined models with a global response alongside group-specific deviations: global and group-specific common-shape model and global and group-specific different-shapes model. All models were fitted using the gam function with beta family (betar, logit link) in the mgcv package⁷⁰. Predictors were modelled with thin-plate regression splines ($k = 10$). For functional-group-specific effects, we applied factor smooths (bs = “fs”) to obtain separate estimates for each group. Smoothing parameters were estimated by restricted maximum likelihood, with a shrinkage penalty to allow splines to collapse to a straight line when the effective degrees of freedom approached one⁷⁰. Model diagnostics were checked with the gam.check function to ensure no systematic patterns in the residuals.

To assess whether latitude predicts variation in interaction specialization, we implemented a multistage modelling approach (Fig. 2j–l). We tested whether latitude had a significant effect on each specialization metric: that is, whether any LSG could be identified. We then visually compared the fitted relationships among models with a significant latitude effect to determine which of the possible LSGs best described each metric and dataset: increasing or decreasing specialization towards the Equator, a general but complex nonlinear pattern or heterogeneous functional-group-specific latitudinal patterns (Fig. 1). First, we evaluated potential methodological biases that may complicate interpretation of previously published results by testing effects of three network descriptors (sampling focus, habitat openness and geographic region), each as a fixed effect in separate models alongside latitude, for each specialization metric and network dataset. Second, network descriptors with significant effects on latitudinal specialization patterns were incorporated as random effects into the main latitudinal models (Fig. 2j). We then re-evaluated each random effect and sequentially removed those that did not remain significant. This resulted in the final set of eight models with latitude as the key predictor and only network descriptors that significantly influenced the LSG retained as random effects (Supplementary Table 4). This multistage

approach accounts for dataset-specific variability in methodological biases while isolating the general latitudinal trend in specialization. Because pooling all network types in the all-networks dataset can generate artefactual patterns due to strong heterogeneity in taxonomic scope, we used the all-networks dataset only to document how pooled analyses compare with earlier syntheses and do not interpret these results further (Extended Data Fig. 2).

To further evaluate potential predictors of the LSG, we tested the importance of climatic, environmental and biotic variables (Fig. 2I and Supplementary Table 2). We first calculated pairwise correlations among all variables (Spearman's ρ ; Fig. 2), visualized their latitudinal distributions (Extended Data Fig. 3) and removed highly correlated predictors ($r > 0.7$) to reduce redundancy. Because elevational variation can disrupt latitudinal patterns, we also tested the relationship of MAT to elevation, latitude and their interaction using a linear model (lm function). MAT was strongly predicted by this model ($P < 0.001$; $R^2 = 0.89$; Extended Data Fig. 4), and we therefore excluded elevation from subsequent analyses. The strong collinearity between latitude and temperature indicates that these variables capture largely overlapping environmental gradients. Accordingly, comparisons of model fit (AIC) are interpreted as reflecting relative explanatory performance rather than fully independent effects. Among the biotic variables, we excluded predicted species richness for birds, nectarivorous birds and butterflies, as these predictors showed very limited variability within the network datasets, probably because of geographical clustering of datasets containing these functional groups, making them unsuitable for robust model testing.

We then evaluated the remaining climatic (MAT, SD_MMT, MAP), environmental (NDVI and SD_NDVI) and biotic (1 predicted and 11 observed species-richness measures) predictors. Because biotic predictors are meaningful at different metric–dataset scales, we deployed them selectively: observed plant species richness was considered broadly for all metrics and datasets; predicted plant species richness, observed pollinator species richness and observed total species richness were considered for H_2' and d'_{plants} in the all-networks and full-networks datasets; and observed richness of functional pollinator groups was used for species-level metrics (d'_{plants} , $d'_{\text{pollinators}}$) across the datasets and additionally for H_2' in the functional-group dataset (see Supplementary Table 6 for details). Following the procedure for the latitudinal models, we selected relevant methodological biases as random effects for each model and retained only those with significant effects. Consequently, each predictor was tested in a separate model, with MAT and MAP also examined as interaction terms. Within each of the eight model sets, we compared models by AIC against the corresponding latitudinal model, treated as a baseline. AIC values were standardized within each model set by setting the latitudinal model to zero, and we report ΔAIC values relative to this baseline (Supplementary Table 7). To compare predictor contributions across models with differing random-effect structure, we also quantified the marginal effect of each predictor. For each fitted model, we constructed a corresponding baseline model containing the same significant random effects but no focal predictor. We then quantified the marginal contribution of the predictor (or MAT $\times$ MAP interaction) as the increase in explained variance (R^2) between the full model and its corresponding random-effects baseline model. This procedure isolates the variation uniquely attributable to the focal predictor while controlling for variation associated with methodological biases (Supplementary Table 7).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data used for modelling (all network characteristics, including specialization metrics, together with all explanatory variables and

other metadata) are available via Zenodo at <https://doi.org/10.5281/zenodo.19118146> (ref. 71). Note that raw interaction matrices are not publicly available because they are often part of ongoing projects. Requests for access to these data should be directed to S.P.S. or R. Tropek, who will coordinate the request with the contributing authors, aiming to respond within a month. The contributing authors will collectively make access decisions.

Code availability

Complete R code used for this analysis is available via Zenodo at <https://doi.org/10.5281/zenodo.19118146> (ref. 71).

References

- Armbruster, W. S. & Muchhala, N. Associations between floral specialization and species diversity: cause, effect, or correlation? *Evol. Ecol.* **23**, 159–179 (2009).
- Van Der Niet, T. & Johnson, S. D. Phylogenetic evidence for pollinator-driven diversification of angiosperms. *Trends Ecol. Evol.* **27**, 353–361 (2012).
- Ollerton, J. Pollinator diversity: distribution, ecological function, and conservation. *Annu. Rev. Ecol. Syst.* **48**, 353–376 (2017).
- Kay, K. M. & Sargent, R. D. The role of animal pollination in plant speciation: integrating ecology, geography, and genetics. *Annu. Rev. Ecol. Syst.* **40**, 637–656 (2009).
- Pinheiro, R. B. P., Felix, G. M. F., Bell, J. A. & Fecchio, A. The latitudinal specialization gradient of bird–malaria parasite networks in South America: lower connectance, but more evenly distributed interactions towards the equator. *Ecography* **2024**, e06763 10.1111/ecog.06763 (2024).
- Hargreaves, A. L. Geographic gradients in species interactions: from latitudinal patterns to ecological mechanisms. *Annu. Rev. Ecol. Syst.* **55**, 369–393 (2024).
- Schleuning, M. et al. Specialization of mutualistic interaction networks decreases toward tropical latitudes. *Curr. Biol.* **22**, 1925–1931 (2012).
- Moles, A. T. & Ollerton, J. Is the notion that species interactions are stronger and more specialized in the tropics a zombie idea? *Biotropica* **48**, 141–145 (2016).
- Dobzhansky, T. Evolution in the tropics. *Am. Sci.* **38**, 208–221 (1950).
- MacArthur, R. H. *Geographical Ecology: Patterns in the Distribution of Species* (Harper & Row, 1972).
- Schemske, D. W., Mittelbach, G. G., Cornell, H. V., Sobel, J. M. & Roy, K. Is there a latitudinal gradient in the importance of biotic interactions? *Annu. Rev. Ecol. Syst.* **40**, 245–269 (2009).
- Pauw, A. Can pollination niches facilitate plant coexistence? *Trends Ecol. Evol.* **28**, 30–37 (2013).
- Brown, J. H. Why are there so many species in the tropics? *J. Biogeogr.* **41**, 8–22 (2014).
- Vázquez, D. P. & Stevens, R. D. The latitudinal gradient in niche breadth: concepts and evidence. *Am. Nat.* **164**, E1–E19 (2004).
- Luna, P., Villalobos, F., Escobar, F., Neves, F. S. & Dáttilo, W. Global trends in the trophic specialisation of flower–visitor networks are explained by current and historical climate. *Ecol. Lett.* **25**, 113–124 (2022).
- Ollerton, J. & Cranmer, L. Latitudinal trends in plant–pollinator interactions: are tropical plants more specialised? *Oikos* **98**, 340–350 (2002).
- Rahimi, E. & Jung, C. Assessing the relationship between latitude and plant–pollinator network specialization. *J. Ecol. Environ.* **48**, 256–262 (2024).
- Dalsgaard, B. et al. Specialization in plant–hummingbird networks is associated with species richness, contemporary precipitation and quaternary climate-change velocity. *PLOS One* **6**, e25891 (2011).
- Trøjelsgaard, K. & Olesen, J. M. Macroecology of pollination networks. *Glob. Ecol. Biogeogr.* **22**, 149–162 (2013).

20. Vizentin-Bugoni, J. et al. Plant-pollinator networks in the tropics: a review. In *Ecological Networks in the Tropics* (eds Dáttilo, W. & Rico-Gray, V.) 73–91 (Springer, 2018).
21. Pauw, A. & Stanway, R. Unrivalled specialization in a pollination network from South Africa reveals that specialization increases with latitude only in the Southern Hemisphere. *J. Biogeogr.* **42**, 652–661 (2015).
22. Cirtwill, A. R. et al. The latitudinal biotic interaction hypothesis revisited: contrasting latitudinal richness gradients in actively vs. passively accumulated interaction partners of honey bees. *BMC Ecol. Evol.* **25**, 24 (2025).
23. Gorostiague, P., Ollerton, J. & Ortega-Baes, P. Latitudinal gradients in biotic interactions: are cacti pollination systems more specialized in the tropics? *Plant Biol.* **25**, 187–191 (2023).
24. Olesen, J. M. & Jordano, P. Geographic patterns in plant-pollinator mutualistic networks. *Ecology* **83**, 2416–2424 (2002).
25. Ollerton, J., Killick, A., Lamborn, E., Watts, S. & Whiston, M. Multiple meanings and modes: on the many ways to be a generalist flower. *Taxon* **56**, 717–728 (2007).
26. Armbruster, W. S. The specialization continuum in pollination systems: diversity of concepts and implications for ecology, evolution and conservation. *Funct. Ecol.* **31**, 88–100 (2017).
27. Doré, M., Fontaine, C. & Thébault, E. Relative effects of anthropogenic pressures, climate, and sampling design on the structure of pollination networks at the global scale. *Glob. Change Biol.* **27**, 1266–1280 (2021).
28. Brimacombe, C., Bodner, K. & Fortin, M.-J. How network size strongly determines trophic specialisation: a technical comment on Luna et al. (2022). *Ecol. Lett.* **25**, 1914–1916 (2022).
29. Classen, A. et al. Specialization of plant–pollinator interactions increases with temperature at Mt. Kilimanjaro. *Ecol. Evol.* **10**, 2182–2195 (2020).
30. Bailey, S. et al. Primary productivity and species richness: relationships among functional guilds, residency groups and vagility classes at multiple spatial scales. *Ecography* **27**, 207–217 (2004).
31. Vázquez, D. P., Blüthgen, N., Cagnolo, L. & Chacoff, N. P. Uniting pattern and process in plant-animal mutualistic networks: a review. *Ann. Bot.* **103**, 1445–1457 (2009).
32. Orr, M. C. et al. Global patterns and drivers of bee distribution. *Curr. Biol.* **31**, 451–458 (2021).
33. Elberling, H. & Olesen, J. M. The structure of a high latitude plant-flower visitor system: the dominance of flies. *Ecography* **22**, 314–323 (1999).
34. Brown, J. H., Gillooly, J. F., Allen, A. P., Savage, V. M. & West, G. B. Toward a metabolic theory of ecology. *Ecology* **85**, 1771–1789 (2004).
35. Petanidou, T. et al. Climate drives plant–pollinator interactions even along small-scale climate gradients: the case of the Aegean. *Plant Biol.* **20**, 176–183 (2018).
36. Fenner, M. The phenology of growth and reproduction in plants. *Perspect. Plant Ecol. Evol. Syst.* **1/1**, 78–91 (1998).
37. Rafferty, N. E., CaraDonna, P. J. & Bronstein, J. L. Phenological shifts and the fate of mutualisms. *Oikos* **124**, 14–21 (2015).
38. Song, B. et al. Global analysis of floral longevity reveals latitudinal gradients and biotic and abiotic correlates. *N. Phytol.* **235**, 2054–2065 (2022).
39. Dzekashu, F. F. et al. Seasonal and elevational changes of plant-pollinator interaction networks in East African mountains. *Ecol. Evol.* **13**, e10060 (2023).
40. Glaum, P., Wood, T. J., Morris, J. R. & Valdovinos, F. S. Phenology and flowering overlap drive specialisation in plant–pollinator networks. *Ecol. Lett.* **24**, 2648–2659 (2021).
41. Maicher, V. et al. Flying between raindrops: strong seasonal turnover of several Lepidoptera groups in lowland rainforests of Mount Cameroon. *Ecol. Evol.* **8**, 12761–12772 (2018).
42. Aizen, M. A. Down-facing flowers, hummingbirds and rain. *Taxon* **52**, 675–680 (2003).
43. Klomberg, Y. et al. Spatiotemporal variation in the role of floral traits in shaping tropical plant-pollinator interactions. *Ecol. Lett.* **25**, 839–850 (2022).
44. Lefebvre, V., Villemant, C., Fontaine, C. & Daugeron, C. Altitudinal, temporal and trophic partitioning of flower-visitors in Alpine communities. *Sci. Rep.* **8**, 4706 (2018).
45. Saunders, M. E. et al. Climate mediates roles of pollinator species in plant–pollinator networks. *Glob. Ecol. Biogeogr.* **32**, 511–518 (2023).
46. Chmel, K. et al. Bird pollination syndrome is the plant's adaptation to ornithophily, but nectarivorous birds are not so selective. *Oikos* **130**, 1411–1424 (2021).
47. Willmer, P. *Pollination and Floral Ecology* (Princeton Univ. Press, 2011).
48. Bartomeus, I. et al. Climate-associated phenological advances in bee pollinators and bee-pollinated plants. *Proc. Natl Acad. Sci. USA* **108**, 20645–20649 (2011).
49. Biella, P., Akter, A., Ollerton, J., Nielsen, A. & Klecka, J. An empirical attack tolerance test alters the structure and species richness of plant–pollinator networks. *Funct. Ecol.* **34**, 2246–2258 (2020).
50. Kudo, G. & Ida, T. Y. Early onset of spring increases the phenological mismatch between plants and pollinators. *Ecology* **94**, 2311–2320 (2013).
51. R Core Team. *R: A Language and Environment for Statistical Computing* (R Foundation for Statistical Computing, 2025).
52. Poisot, T. et al. mangal – making ecological network analysis simple. *Ecography* **39**, 384–390 (2016).
53. Fortuna, M. A., Ortega, R. & Bascompte, J. The web of life. Preprint at <https://doi.org/10.48550/arXiv.1403.2575> (2014).
54. Butterill, P., Jorge, L., Xing, S. & Fayle, T. LifeWebs: a (global) database of bipartite ecological interaction networks. *Biodivers. Inf. Sci. Stand.* **5**, e75626 (2021).
55. Hijmans, R. J. et al. terra: spatial data analysis. R package version 1.9-25 (2025).
56. Fick, S. E. & Hijmans, R. J. WorldClim 2: new 1-km spatial resolution climate surfaces for global land areas. *Int. J. Climatol.* **37**, 4302–4315 (2017).
57. Didan, K. MODIS/terra vegetation indices 16-day L3 global 250m SIN Grid V006. NASA <https://doi.org/10.5067/MODIS/MOD13Q1.006> (2015).
58. Cai, L. et al. Global models and predictions of plant diversity based on advanced machine learning techniques. *N. Phytol.* **237**, 1432–1445 (2023).
59. Jenkins, C. N., Pimm, S. L. & Joppa, L. N. Global patterns of terrestrial vertebrate diversity and conservation. *Proc. Natl Acad. Sci. USA* **110**, E2602–E2610 (2013).
60. Daru, B. H. A global database of butterfly species native distributions. *Ecology* **106**, e4462 <https://doi.org/10.1002/ecy.4462> (2025).
61. Hijmans, R. J. et al. raster: geographic data analysis and modeling. R package version 3.6-32 (2025).
62. Chamberlain, S. A. & Szöcs, E. taxize: taxonomic search and retrieval in R. *F1000Res.* **2**, 191 (2013).
63. GBIF Secretariat. GBIF backbone taxonomy. GBIF <https://doi.org/10.15468/39OMEI> (2023).
64. Ollerton, J. *Pollinators & Pollination: Nature and Society* (Pelagic, 2021).
65. Dormann, C. F., Gruber, B. & Fründ, J. Introducing the bipartite package: analysing ecological networks. *R J.* **8**, 8–11 (2008).
66. Blüthgen, N., Menzel, F. & Blüthgen, N. Measuring specialization in species interaction networks. *BMC Ecol.* **6**, 9 (2006).
67. Blüthgen, N. & Staab, M. A critical evaluation of network approaches for studying species interactions. *Annu. Rev. Ecol. Evol. Syst.* **55**, 65–88 (2024).

68. Chiu, C.-H., Chao, A., Vogel, S., Kriegel, P. & Thorn, S. Quantifying and estimating ecological network diversity based on incomplete sampling data. *Philos. Trans. R. Soc. B* **378**, 20220183 (2023).
69. Pedersen, E. J., Miller, D. L., Simpson, G. L. & Ross, N. Hierarchical generalized additive models in ecology: an introduction with mgcv. *PeerJ* **7**, e6876 (2019).
70. Wood, S. N. Fast stable restricted maximum likelihood and marginal likelihood estimation of semiparametric generalized linear models. *J. R. Stat. Soc. B* **73**, 3–36 (2011).
71. Sakhalkar, S. P. et al. Modelling data and codebase for Sakhalkar, S.P. et al., (2026). Global patterns in plant–pollinator specialisation. *Nature Ecology & Evolution* (accepted). *Zenodo* <https://doi.org/10.5281/zenodo.19118146> (2026).
72. Pianka, E. R. Latitudinal gradients in species diversity: a review of concepts. *Am. Nat.* **100**, 33–46 (1966).

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Author contributions

S.P.S. and R. Tropek conceived and designed the study, with substantial input from N.B.; S.P.S. collated and processed the datasets and performed the analyses under the supervision of R. Tropek, with feedback from N.B., B.D., C.F.D., C.N.K.-B., D.P.V., J.O., J.R., L.A.B., M.S., P.C. and T.M.K. S.P.S. and R. Tropek prepared all visualizations, tables and supplementary materials, with contributions from D.A., J.F. and G.U.-G. S.P.S. and R. Tropek wrote the manuscript draft, with contributions from N.B., B.D., C.F.D., C.N.K.-B., D.P.V., J.O., J.R., L.A.B., M.S., P.C., T.M.K. and T.M.F. All authors contributed data, critically reviewed the manuscript and approved the final version for submission.

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Competing interests

The authors declare no competing interests.

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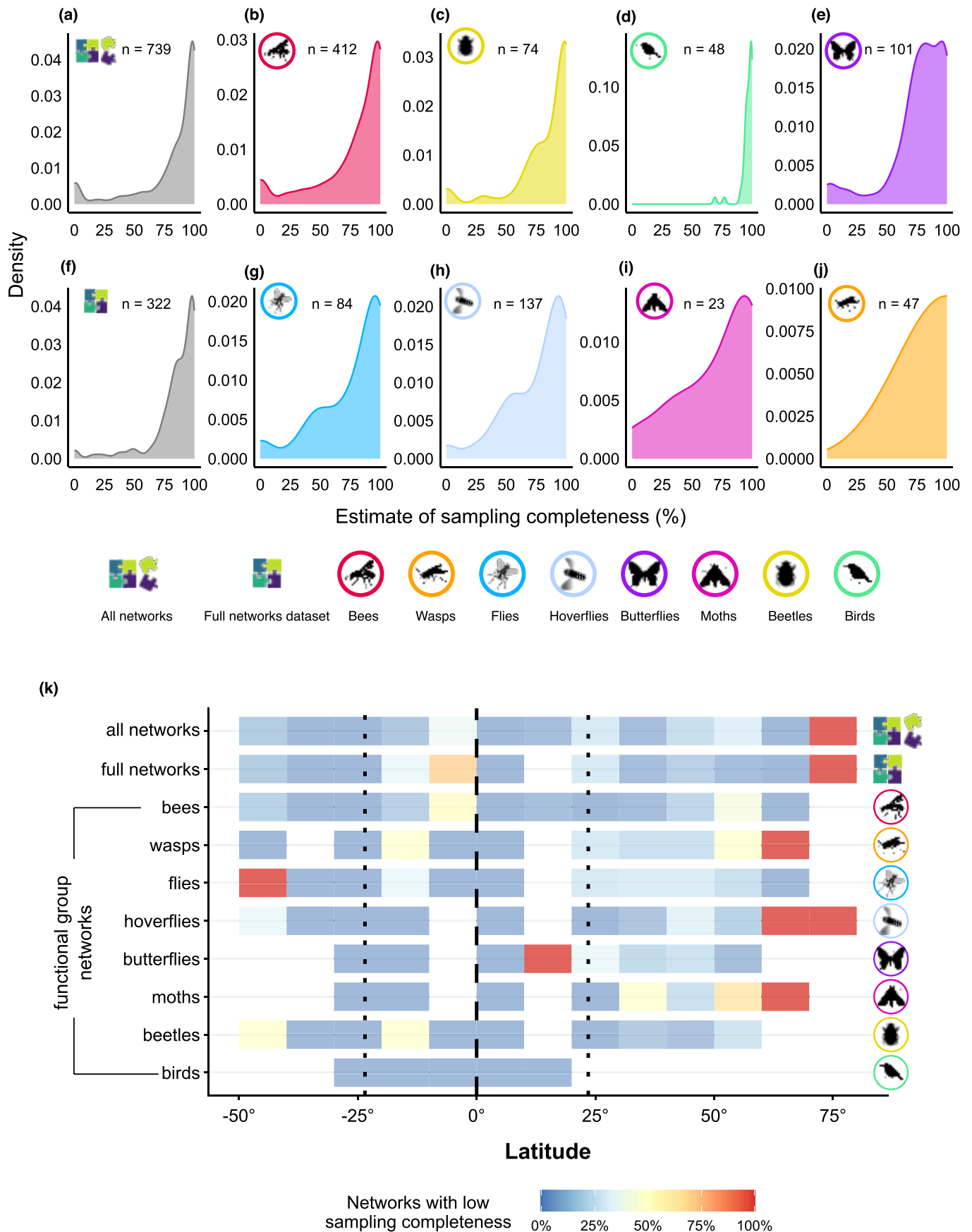
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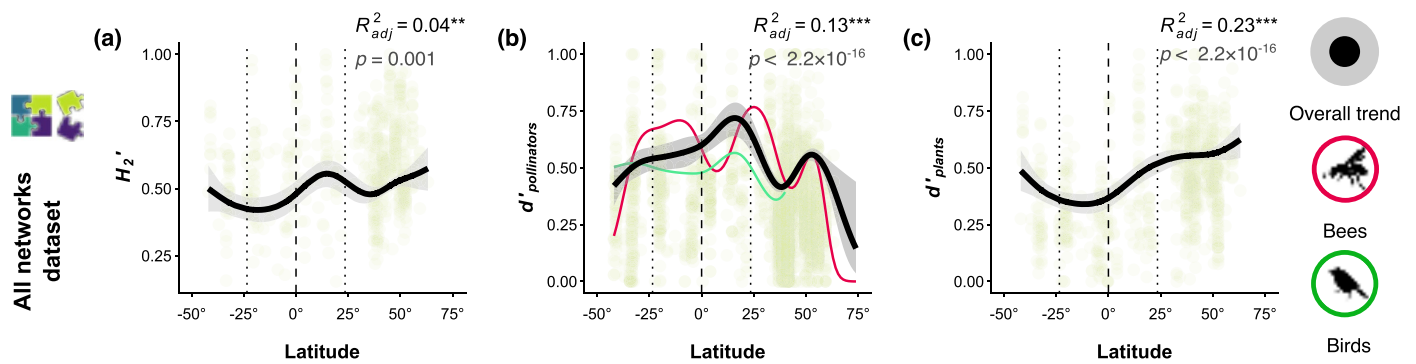
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Extended Data Fig. 1 | See next page for caption.

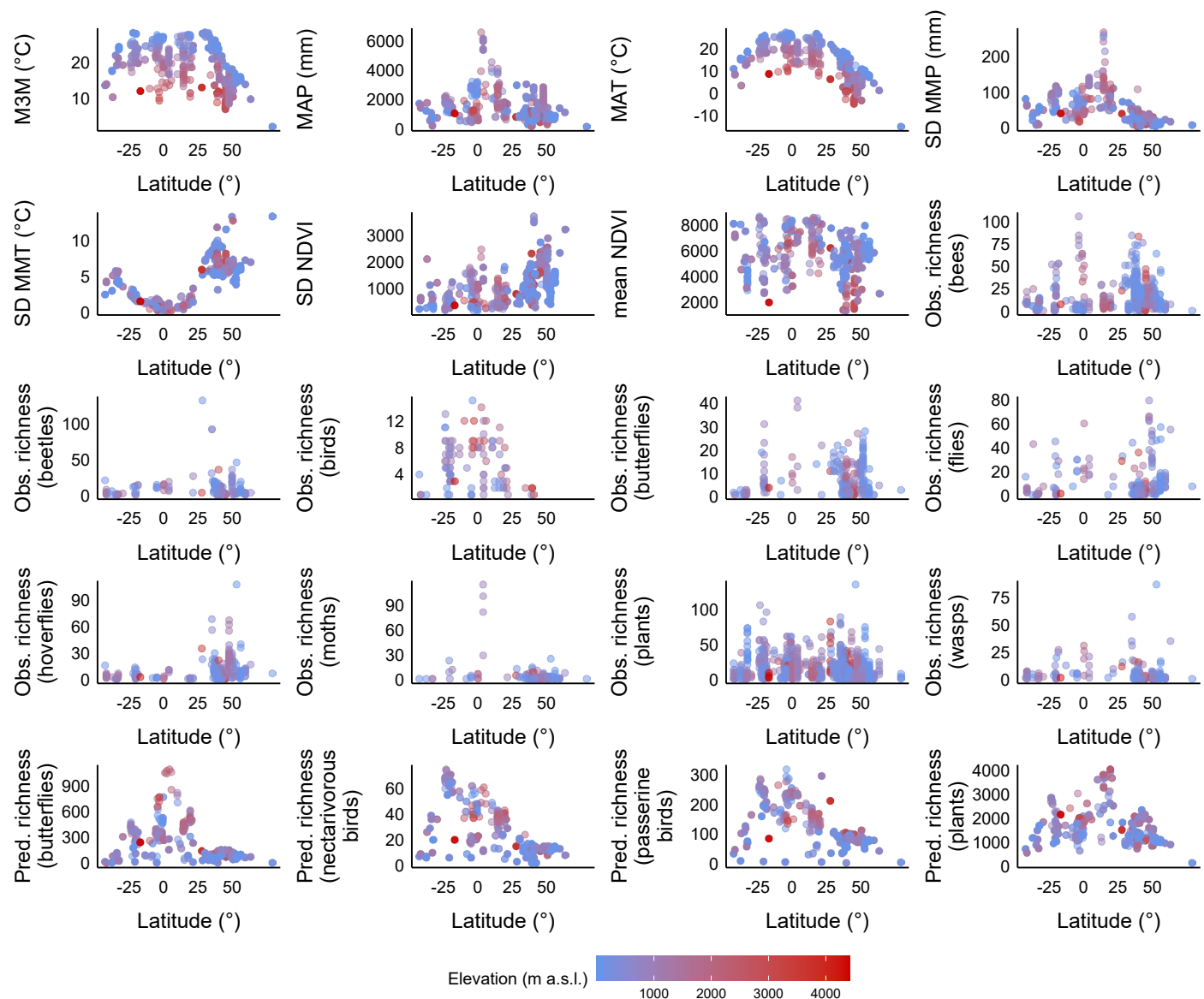
Extended Data Fig. 1 | Sampling completeness of the analysed networks. (a–j) Kernel density distributions of sampling completeness (%) for the all networks dataset (a), the full networks dataset (f), and individual functional groups in the functional-group networks dataset (b–e, g–j); panel labels report sample sizes (n). (k) Latitudinal distribution of low-completeness networks,

shown as the proportion of networks with sampling completeness ≤ 0.7 in 10° latitudinal bins for each dataset and functional group in the functional-group networks dataset. Credit: Silhouettes from PhyloPic (<https://www.phylopic.org/>) under a Creative Commons license: *Halictus rubicundus* (CC BY 4.0); *Acanthorhynchus tenuirostris* (CC0 1.0).



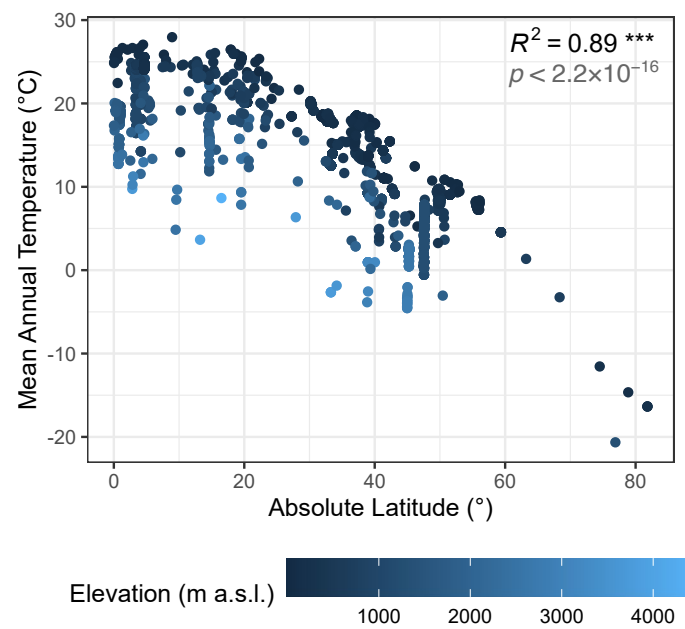
Extended Data Fig. 2 | Latitudinal patterns in plant–pollinator specialisation metrics for the all networks dataset. Panels show specialisation metrics: network-level H_2' (a); and community-mean species-level separated for pollinators ($d'_{\text{pollinators}}$; b) and for plants (d'_{plants} ; c). Vertical dashed line marks the equator (0°), and vertical dotted lines mark the Tropics of Capricorn (-23.5°) and Cancer (23.5°). Curves are GAMM/HGAM fitted mean response smooths: black curves indicate overall latitudinal trends with 95% confidence intervals (grey shading), coloured curves represent significant functional group-specific trends (see legend). Coloured points are raw data. Curves are drawn only when

the smooth term for latitude is significant, otherwise only points are shown. Adjusted R^2 and exact p-values for the overall trend are reported in each panel (see Table S5 for full model summaries). We assessed the significance of smooth terms using two-sided approximate F-tests without correction for multiple comparisons. Credit: Silhouettes from PhyloPic (<https://www.phylopic.org/>) under a Creative Commons license: *Onthophagus taurus*, *Darapsa myron* (CC BY 3.0); *Acanthorhynchus tenuirostris*, *Papilio machaon*, Tachinidae, *Eupeodes volucris*, *Vespula germanica* (CC0 1.0).



Extended Data Fig. 3 | Latitudinal distribution of plant–pollinator network descriptors. Scatterplots show the values of climatic, environmental, and biotic descriptors for individual networks in the all networks dataset plotted against latitude; points are coloured by elevation. Descriptor definitions, abbreviations

and units are given in Table S2. Credit: Silhouette from PhyloPic (<https://www.phylopic.org/>) under a Creative Commons license: *Acanthorhynchus tenuirostris* (CC01.0).



Extended Data Fig. 4 | Mean annual temperature as a function of latitude and elevation for the analysed plant-pollinator interaction networks. Sampling sites are coloured by their elevation. The reported adjusted R^2 and significance (*** $p < 0.001$) originate from a linear model predicting mean annual temperature from absolute latitude and elevation ($F_{3,3492} = 8879$).

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| ☐ | ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☐ | ☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☐ | ☒ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☐ | ☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	No software was used for data collection.
Data analysis	The codebase for this manuscript is available at Zenodo (https://zenodo.org/records/19118146; DOI: https://doi.org/10.5281/zenodo.19118146). The analysis was conducted entirely in R, for which the following packages were used: - Data processing and visualisation: dbplyr v 2.5.0; gtExtras v 0.5.0; gt v 1.0.0; patchwork v 1.3.1; taxize v 0.10.0; tidyverse v 2.0.0 - Metric calculations: bipartite v 2.21 - Sampling completeness estimates: iNEXT.link v 3.0.2 - Modelling: mgcv 1.9; tidymodels v 1.3.0- Spatial data extraction: raster v 3.6.3-2; terra v 1.8.4-2 - Interactive app: leaflet v 2.2.2; shiny v 1.10.0

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

All data used for statistical analyses (all network characteristics, including specialisation metrics, together with all explanatory variables and other metadata) is made available on Zenodo (<https://doi.org/10.5281/zenodo.19118146>). Note that raw interaction matrices are not publicly available because they are often part of ongoing projects. Requests for access to these data should be directed to Sailee P. Sakhalkar (sailee.sakha@gmail.com) or Robert Tropek (robert.tropek@gmail.com), who will coordinate the request with the contributing authors, aiming to revert in a month's time. The contributing authors will collectively make access decisions.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	NA
Reporting on race, ethnicity, or other socially relevant groupings	NA
Population characteristics	NA
Recruitment	NA
Ethics oversight	NA

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☐ Behavioural & social sciences ☒ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Describe how sample size was determined, detailing any statistical methods used to predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient.
Data exclusions	Describe any data exclusions. If no data were excluded from the analyses, state so OR if data were excluded, describe the exclusions and the rationale behind them, indicating whether exclusion criteria were pre-established.
Replication	Describe the measures taken to verify the reproducibility of the experimental findings. If all attempts at replication were successful, confirm this OR if there are any findings that were not replicated or cannot be reproduced, note this and describe why.
Randomization	Describe how samples/organisms/participants were allocated into experimental groups. If allocation was not random, describe how covariates were controlled OR if this is not relevant to your study, explain why.
Blinding	Describe whether the investigators were blinded to group allocation during data collection and/or analysis. If blinding was not possible, describe why OR explain why blinding was not relevant to your study.

Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	Briefly describe the study type including whether data are quantitative, qualitative, or mixed-methods (e.g. qualitative cross-sectional,
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Study description	<i>quantitative experimental, mixed-methods case study).</i>
Research sample	<i>State the research sample (e.g. Harvard university undergraduates, villagers in rural India) and provide relevant demographic information (e.g. age, sex) and indicate whether the sample is representative. Provide a rationale for the study sample chosen. For studies involving existing datasets, please describe the dataset and source.</i>
Sampling strategy	<i>Describe the sampling procedure (e.g. random, snowball, stratified, convenience). Describe the statistical methods that were used to predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient. For qualitative data, please indicate whether data saturation was considered, and what criteria were used to decide that no further sampling was needed.</i>
Data collection	<i>Provide details about the data collection procedure, including the instruments or devices used to record the data (e.g. pen and paper, computer, eye tracker, video or audio equipment) whether anyone was present besides the participant(s) and the researcher, and whether the researcher was blind to experimental condition and/or the study hypothesis during data collection.</i>
Timing	<i>Indicate the start and stop dates of data collection. If there is a gap between collection periods, state the dates for each sample cohort.</i>
Data exclusions	<i>If no data were excluded from the analyses, state so OR if data were excluded, provide the exact number of exclusions and the rationale behind them, indicating whether exclusion criteria were pre-established.</i>
Non-participation	<i>State how many participants dropped out/declined participation and the reason(s) given OR provide response rate OR state that no participants dropped out/declined participation.</i>
Randomization	<i>If participants were not allocated into experimental groups, state so OR describe how participants were allocated to groups, and if allocation was not random, describe how covariates were controlled.</i>

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	We analysed latitudinal patterns in plant–pollinator interactions using an unprecedented global dataset of quantitative plant–pollinator networks. We quantified specialisation with three complementary metrics at the network- and species- levels, separately for plants, pollinators, and major pollinator functional groups, and related these to climatic and diversity variables to directly test classical hypotheses about the drivers of latitudinal specialisation. All analyses explicitly accounted for sampling completeness and other potential biases that affected previous, smaller-scale studies, yielding robust and comparable results across regions and taxa.
Research sample	We collated published and unpublished studies on plant–pollinator interactions to assemble >3,400 quantitative networks comprising >110,000 interactions, spanning all major terrestrial biomes and pollinator taxa.
Sampling strategy	We compiled the global dataset through a systematic search of large bibliographic databases, complemented by forward–backward citation screening. We additionally incorporated datasets from three existing repositories/databases. Finally, we contacted the authors of studies that did not have openly accessible data, and invited contributions of both published and unpublished datasets via professional networks and a targeted email campaign.
Data collection	Data were collected by the authors of the original studies (including all co-authors of this manuscript) using established field protocols for quantitative interaction sampling. Basic details for each contributing dataset are provided in Table S1.
Timing and spatial scale	The dataset includes quantitative networks sampled between the 1960s and 2023 and covers virtually all terrestrial biogeographic regions and major ecosystem types.
Data exclusions	We included plant–pollinator networks that : (i) originated from natural or semi-natural ecosystems (excluding agricultural, urban and other strongly anthropogenic habitats); (ii) contained at least five plant and five pollinator species; (iii) reported directly sampled interactions (excluding broad spatio-temporal compilations); and (iv) provided quantitative interaction strengths (excluding binary matrices). For analyses, we defined three datasets: All networks (no exclusions beyond the criteria above), Full networks (taxonomically unbiased networks), and Functional-group networks (taxonomically biased networks and subnetworks used to resolve functional-group patterns).
Reproducibility	The codebase (with scripts and data) for this manuscript is available at Zenodo (https://zenodo.org/records/19118146 ; DOI: https://doi.org/10.5281/zenodo.19118146).
Randomization	Not applicable. We analysed all eligible networks that met our inclusion criteria to characterise global patterns
Blinding	Not applicable. Analyses required exact geographic coordinates and study identities to quantify spatial patterns and environmental associations.
Did the study involve field work?	☐ Yes ☒ No

Field work, collection and transport

Field conditions	<i>Describe the study conditions for field work, providing relevant parameters (e.g. temperature, rainfall).</i>
Location	<i>State the location of the sampling or experiment, providing relevant parameters (e.g. latitude and longitude, elevation, water depth).</i>
Access & import/export	<i>Describe the efforts you have made to access habitats and to collect and import/export your samples in a responsible manner and in compliance with local, national and international laws, noting any permits that were obtained (give the name of the issuing authority, the date of issue, and any identifying information).</i>
Disturbance	<i>Describe any disturbance caused by the study and how it was minimized.</i>

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	<i>Describe all antibodies used in the study; as applicable, provide supplier name, catalog number, clone name, and lot number.</i>
Validation	<i>Describe the validation of each primary antibody for the species and application, noting any validation statements on the manufacturer's website, relevant citations, antibody profiles in online databases, or data provided in the manuscript.</i>

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)	<i>State the source of each cell line used and the sex of all primary cell lines and cells derived from human participants or vertebrate models.</i>
Authentication	<i>Describe the authentication procedures for each cell line used OR declare that none of the cell lines used were authenticated.</i>
Mycoplasma contamination	<i>Confirm that all cell lines tested negative for mycoplasma contamination OR describe the results of the testing for mycoplasma contamination OR declare that the cell lines were not tested for mycoplasma contamination.</i>
Commonly misidentified lines (See ICLAC register)	<i>Name any commonly misidentified cell lines used in the study and provide a rationale for their use.</i>

Palaeontology and Archaeology

Specimen provenance	<i>Provide provenance information for specimens and describe permits that were obtained for the work (including the name of the issuing authority, the date of issue, and any identifying information). Permits should encompass collection and, where applicable, export.</i>
Specimen deposition	<i>Indicate where the specimens have been deposited to permit free access by other researchers.</i>

Dating methods

If new dates are provided, describe how they were obtained (e.g. collection, storage, sample pretreatment and measurement), where they were obtained (i.e. lab name), the calibration program and the protocol for quality assurance OR state that no new dates are provided.

☐ Tick this box to confirm that the raw and calibrated dates are available in the paper or in Supplementary Information.

Ethics oversight

Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Animals and other research organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research, and [Sex and Gender in Research](#)

Laboratory animals

For laboratory animals, report species, strain and age OR state that the study did not involve laboratory animals.

Wild animals

Provide details on animals observed in or captured in the field; report species and age where possible. Describe how animals were caught and transported and what happened to captive animals after the study (if killed, explain why and describe method; if released, say where and when) OR state that the study did not involve wild animals.

Reporting on sex

Indicate if findings apply to only one sex; describe whether sex was considered in study design, methods used for assigning sex. Provide data disaggregated for sex where this information has been collected in the source data as appropriate; provide overall numbers in this Reporting Summary. Please state if this information has not been collected. Report sex-based analyses where performed, justify reasons for lack of sex-based analysis.

Field-collected samples

For laboratory work with field-collected samples, describe all relevant parameters such as housing, maintenance, temperature, photoperiod and end-of-experiment protocol OR state that the study did not involve samples collected from the field.

Ethics oversight

Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Clinical data

Policy information about [clinical studies](#)

All manuscripts should comply with the ICMJE [guidelines for publication of clinical research](#) and a completed [CONSORT checklist](#) must be included with all submissions.

Clinical trial registration

Provide the trial registration number from ClinicalTrials.gov or an equivalent agency.

Study protocol

Note where the full trial protocol can be accessed OR if not available, explain why.

Data collection

Describe the settings and locales of data collection, noting the time periods of recruitment and data collection.

Outcomes

Describe how you pre-defined primary and secondary outcome measures and how you assessed these measures.

Dual use research of concern

Policy information about [dual use research of concern](#)

Hazards

Could the accidental, deliberate or reckless misuse of agents or technologies generated in the work, or the application of information presented in the manuscript, pose a threat to:

No	Yes
☐	☐ Public health
☐	☐ National security
☐	☐ Crops and/or livestock
☐	☐ Ecosystems
☐	☐ Any other significant area

Experiments of concern

Does the work involve any of these experiments of concern:

No	Yes
☐	☐ Demonstrate how to render a vaccine ineffective
☐	☐ Confer resistance to therapeutically useful antibiotics or antiviral agents
☐	☐ Enhance the virulence of a pathogen or render a nonpathogen virulent
☐	☐ Increase transmissibility of a pathogen
☐	☐ Alter the host range of a pathogen
☐	☐ Enable evasion of diagnostic/detection modalities
☐	☐ Enable the weaponization of a biological agent or toxin
☐	☐ Any other potentially harmful combination of experiments and agents

Plants

Seed stocks	NA
Novel plant genotypes	NA
Authentication	NA

ChIP-seq

Data deposition

- ☐ Confirm that both raw and final processed data have been deposited in a public database such as [GEO](#).
- ☐ Confirm that you have deposited or provided access to graph files (e.g. BED files) for the called peaks.

Data access links <i>May remain private before publication.</i>	<i>For "Initial submission" or "Revised version" documents, provide reviewer access links. For your "Final submission" document, provide a link to the deposited data.</i>
Files in database submission	<i>Provide a list of all files available in the database submission.</i>
Genome browser session (e.g. UCSC)	<i>Provide a link to an anonymized genome browser session for "Initial submission" and "Revised version" documents only, to enable peer review. Write "no longer applicable" for "Final submission" documents.</i>

Methodology

Replicates	<i>Describe the experimental replicates, specifying number, type and replicate agreement.</i>
Sequencing depth	<i>Describe the sequencing depth for each experiment, providing the total number of reads, uniquely mapped reads, length of reads and whether they were paired- or single-end.</i>
Antibodies	<i>Describe the antibodies used for the ChIP-seq experiments; as applicable, provide supplier name, catalog number, clone name, and lot number.</i>
Peak calling parameters	<i>Specify the command line program and parameters used for read mapping and peak calling, including the ChIP, control and index files used.</i>
Data quality	<i>Describe the methods used to ensure data quality in full detail, including how many peaks are at FDR 5% and above 5-fold enrichment.</i>
Software	<i>Describe the software used to collect and analyze the ChIP-seq data. For custom code that has been deposited into a community repository, provide accession details.</i>

Flow Cytometry

Plots

Confirm that:

- ☐ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☐ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☐ All plots are contour plots with outliers or pseudocolor plots.
- ☐ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation

Describe the sample preparation, detailing the biological source of the cells and any tissue processing steps used.

Instrument

Identify the instrument used for data collection, specifying make and model number.

Software

Describe the software used to collect and analyze the flow cytometry data. For custom code that has been deposited into a community repository, provide accession details.

Cell population abundance

Describe the abundance of the relevant cell populations within post-sort fractions, providing details on the purity of the samples and how it was determined.

Gating strategy

Describe the gating strategy used for all relevant experiments, specifying the preliminary FSC/SSC gates of the starting cell population, indicating where boundaries between "positive" and "negative" staining cell populations are defined.

- ☐ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.

Magnetic resonance imaging

Experimental design

Design type

Indicate task or resting state; event-related or block design.

Design specifications

Specify the number of blocks, trials or experimental units per session and/or subject, and specify the length of each trial or block (if trials are blocked) and interval between trials.

Behavioral performance measures

State number and/or type of variables recorded (e.g. correct button press, response time) and what statistics were used to establish that the subjects were performing the task as expected (e.g. mean, range, and/or standard deviation across subjects).

Acquisition

Imaging type(s)

Specify: functional, structural, diffusion, perfusion.

Field strength

Specify in Tesla

Sequence & imaging parameters

Specify the pulse sequence type (gradient echo, spin echo, etc.), imaging type (EPI, spiral, etc.), field of view, matrix size, slice thickness, orientation and TE/TR/flip angle.

Area of acquisition

State whether a whole brain scan was used OR define the area of acquisition, describing how the region was determined.

Diffusion MRI

☐ Used

☐ Not used

Preprocessing

Preprocessing software

Provide detail on software version and revision number and on specific parameters (model/functions, brain extraction, segmentation, smoothing kernel size, etc.).

Normalization

If data were normalized/standardized, describe the approach(es): specify linear or non-linear and define image types used for transformation OR indicate that data were not normalized and explain rationale for lack of normalization.

Normalization template

Describe the template used for normalization/transformation, specifying subject space or group standardized space (e.g. original Talairach, MNI305, ICBM152) OR indicate that the data were not normalized.

Noise and artifact removal

Describe your procedure(s) for artifact and structured noise removal, specifying motion parameters, tissue signals and physiological signals (heart rate, respiration).

Volume censoring

Define your software and/or method and criteria for volume censoring, and state the extent of such censoring.

Statistical modeling & inference

Model type and settings

Specify type (mass univariate, multivariate, RSA, predictive, etc.) and describe essential details of the model at the first and second levels (e.g. fixed, random or mixed effects; drift or auto-correlation).

Effect(s) tested

Define precise effect in terms of the task or stimulus conditions instead of psychological concepts and indicate whether ANOVA or factorial designs were used.

Specify type of analysis: ☐ Whole brain ☐ ROI-based ☐ Both

Statistic type for inference

Specify voxel-wise or cluster-wise and report all relevant parameters for cluster-wise methods.

(See [Eklund et al. 2016](#))

Correction

Describe the type of correction and how it is obtained for multiple comparisons (e.g. FWE, FDR, permutation or Monte Carlo).

Models & analysis

n/a | Involved in the study

☐☐ Functional and/or effective connectivity☐☐ Graph analysis☐☐ Multivariate modeling or predictive analysis

Functional and/or effective connectivity

Report the measures of dependence used and the model details (e.g. Pearson correlation, partial correlation, mutual information).

Graph analysis

Report the dependent variable and connectivity measure, specifying weighted graph or binarized graph, subject- or group-level, and the global and/or node summaries used (e.g. clustering coefficient, efficiency, etc.).

Multivariate modeling and predictive analysis

Specify independent variables, features extraction and dimension reduction, model, training and evaluation metrics.